

SPATIALLY-COUPLED OTFS SYSTEMS VIA BLOCK MARKOV SUPERPOSITION TRANSMISSION

Jiayi Yang¹, Shuangyang Li², Qianfan Wang³, Linqi Song^{3,4}, Bo Ai⁵, Xiao Ma¹, and Giuseppe Caire²

¹School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou 510006

²Electrical Engineering and Computer Science, Technical University of Berlin, Berlin, 10587, Germany

³Department of Computer Science, City University of Hong Kong, Hong Kong SAR, China

⁴Shenzhen Research Institute City University of Hong Kong, Shenzhen 518000, China

⁵School of Electronics and Information Engineering, Beijing Jiaotong University, Beijing 100044, China

ABSTRACT

This paper investigates a novel spatially coupled orthogonal time frequency space (SC-OTFS) transmission scheme by integrating block Markov superposition transmission (BMST) with OTFS modulation. We first statistically analyze the distribution of detector mutual information across coded blocks and establish a decoding-threshold-based lower bound to characterize the asymptotic performance of conventional LDPC-coded OTFS systems. The analysis reveals that increasing the code length yields only marginal coding gains in typical doubly-selective channels with limited resolvable paths. Motivated by this observation, we propose an SC-OTFS system in which a portion of each previously coded block is superimposed onto the current block and subsequently modulated using OTFS, thereby exploiting both coding gain and inter-block diversity gain in high-mobility channels. Furthermore, an extrinsic information transfer (EXIT) chart analysis is developed for the SC-OTFS system, which enables optimization of the critical superposition fraction based on decoding thresholds. Simulation results demonstrate that this optimization leads to a significant performance improvement, achieving more than 3 dB gain over conventional 5G LDPC-coded OTFS systems.

Index Terms— 5G, block Markov superposition transmission (BMST), LDPC codes, orthogonal time frequency space (OTFS), high-mobility channel.

1. INTRODUCTION

Orthogonal time frequency space (OTFS) modulation has been proposed in [1] and is now widely recognized as a

promising technique for reliable high-mobility communications in future 6G systems [2]. Unlike OFDM, which operates in the time–frequency (TF) domain, OTFS modulation maps information symbols into the delay–Doppler (DD) domain. Through the inverse symplectic finite Fourier transform (ISFFT), each symbol is spread across the full bandwidth and frame duration, yielding potentially full diversity gain and improved error-correction performance [3].

To ensure reliable transmission, OTFS needs to be combined with powerful channel coding. Prior studies have explored several initial forms of integration. In [4], the performance of coded OTFS was analyzed, highlighting the trade-off between diversity and coding gain. Space–time coding was introduced in OTFS [5], while a low-complexity detection algorithm for Polar-coded OTFS was proposed in [6]. Low-density parity-check (LDPC) codes, adopted in 5G [7], have also been incorporated into OTFS systems, and their error-correction performance has been evaluated in [8]. Despite these efforts, most existing works still adopt a separate design, applying block codes to OTFS without leveraging potential gains from joint design. Moreover, when the channel exhibits only a few resolvable paths, such as in mmWave transmissions [9], the inherent diversity of OTFS may be insufficient, necessitating retransmission. A practical approach to achieve efficient retransmission is block Markov superposition transmission (BMST) [10, 11, 12], which realizes retransmission through finite-field superposition, thereby providing coding gain while preserving spectral efficiency.

In this paper, based on the BMST framework and partial superposition [13, 14, 15, 16, 17], we propose a novel spatially coupled OTFS (SC-OTFS) scheme. The main contributions are summarized as follows: i) **Asymptotic Performance Analysis:** We analyze the distribution of the detector’s mutual information (MI) and derive a decoding-threshold-based lower bound for LDPC-coded OTFS systems. The numerical results show that increasing code length provides only marginal gains for 5G LDPC coded OTFS system over doubly-selective channels with limited resolvable

This work was supported in part by the NSFC under Grants 62301617, 62471506, 62221001, 62341127 and 62371411; by the Guangdong Basic and Applied Basic Research Foundation under Grant 2025A1515011650; by the Young Talent Support Project of the Guangzhou Association for Science and Technology under Grant QT-2025-048; and by the Research Grants Council of the Hong Kong SAR under Grants GRF 11217823 and 11216225. (Corresponding author: Qianfan Wang)

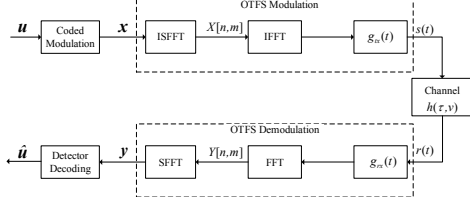


Fig. 1. The block diagram of a coded OTFS system.

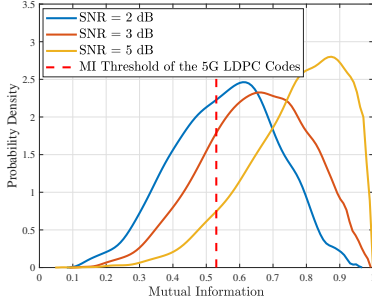


Fig. 2. Estimated PDF of the detector's mutual information for coded OTFS system under various SNRs.

paths. ii) **Novel SC-OTFS Design:** We propose an SC-OTFS system using the BMST manner, where part of each 5G LDPC coded block is superimposed onto the next one before OTFS modulation. This enables both coding gain and inter-block diversity gain in high-mobility channels, with a sliding-window decoding algorithm ensuring low latency. iii) **Design Optimization:** We develop an EXIT chart analysis tailored to SC-OTFS by sampling the mutual information distribution of the OTFS detector and incorporating the 5G LDPC protograph. Based on the decoding thresholds, the superposition fraction is optimized, achieving about a 3 dB gain over conventional 5G LDPC-coded OTFS systems.

2. SPATIALLY COUPLED OTFS SYSTEMS WITH 5G LDPC CODES

2.1. Asymptotic Performance for LDPC Coded OTFS

Consider a coded OTFS system, as shown in Fig. 1. At the transmitter, the information bits u are first encoded by a channel encoder (e.g., LDPC), and then modulated, and reshaped into an $M \times N$ symbol matrix in the delay-Doppler (DD) domain. Through the inverse symplectic finite Fourier transform (ISFFT), the DD-domain symbols are mapped into the time-frequency (TF) domain, and the time-domain signal $s(t)$ is generated via pulse shaping by $g_{tx}(t)$. The wireless channel is modeled in the DD domain as a sum of P resolvable paths with distinct delay and Doppler shifts.

At the receiver, the received signal passed through this channel is filtered and demodulated into TF-domain symbols,

followed by the symplectic finite Fourier transform (SFFT) to obtain DD-domain observations. These noisy DD symbols y are then passed to the detector and channel decoder for data recovery. In this paper, we focus on the sum-product algorithm (SPA) [18, 19, 20] as the OTFS detector. To capture the time-varying nature of high-mobility channels, we evaluate the bit-wise MI between the transmitted bits and the soft log-likelihood ratios (LLRs) generated by the SPA detector [18, 19, 20] across coded blocks, using Monte Carlo simulations. As shown in Fig. 2 for $P = 5$ independent paths, the SPA detector's MI fluctuates significantly, indicating non-ergodic behavior. Although OTFS in principle can achieve full channel diversity through the ISFFT, the demodulated signal effectively experiences significant fluctuated detection quality under such conditions. Based on this observation, we can derive the lower bound of the LDPC-coded OTFS system through decoding threshold analysis. Specifically, for a given protograph or degree distribution of the LDPC code, the decoding threshold can be obtained in terms of SNR, which also corresponds to a threshold mutual information I_{th} . Clearly, when the input SNR exceeds the threshold SNR or equivalently, when the mutual information of the decoder exceeds I_{th} , the LDPC code can converge to vanishing probability of error in the infinite code length limit. Based on this, we can derive the lower bound on frame error rate (FER) for the LDPC coded OTFS system in the infinite code length regime, which can be expressed as

$$P_{FER} \geq \Pr\{I_{E, Det}(\mathbf{X}; \mathbf{Y}) < I_{th}\} = \int_0^{I_{th}} f_{E, Det}(I) dI, \quad (1)$$

where $I_{E, Det}(\mathbf{X}; \mathbf{Y})$ denotes the output MI of the SPA detector and $f_{E, Det}$ denotes the PDF of $I_{E, Det}(\mathbf{X}; \mathbf{Y})$.

Example 1: Consider a 5G LDPC coded OTFS system with a UE speed of 150 km/h and five independent resolvable paths ($P = 5$). We plot the threshold-based lower bound according to Eq. (1), which essentially corresponds to the left-tail probability below $I_{th} \approx 0.53$ for the 5G LDPC codes, as shown in Fig. 2. We further simulate the actual performance of the 5G LDPC coded OTFS system under different code lengths, using LDPC codes constructed from BG2 with lengths $n \in \{512, 1024, 2080\}$ and a code rate $r = 1/2$. The threshold-based lower bound and the simulation results are shown in Fig. 3, from which we observe that, as the code length increases, the incremental coding gain becomes marginal, and the simulated performance gradually converges to the proposed threshold-based lower bound.

The lower-bound analysis indicates that merely increasing the code length provides limited gain. In this case, feedback-based retransmission could improve performance but introduces extra overhead and latency. To address this, we propose SC-OTFS, where part of the current codeword is superimposed onto the next. This allows blocks affected by deep fading to exploit subsequent blocks with better channel conditions, yielding BMST coding gain and inter-block diversity. Even with rapid time-frequency variations, diversity is limited

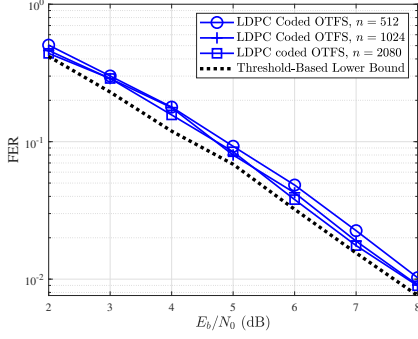


Fig. 3. The threshold-based lower bound and the performance for 5G LDPC coded OTFS with $n \in \{512, 1024, 2080\}$.

by the number of resolvable paths, i.e., the channel behaves as block-fading. This is reflected in the MI distribution of detector and motivates inter-block coupling.

2.2. Transceiver Design of the SC-OTFS Systems

Consider an information sequence $\mathbf{u} = (\mathbf{u}^{(0)}, \dots, \mathbf{u}^{(L-1)})$ of L blocks, where each $\mathbf{u}^{(t)} \in \mathbb{F}_2^k$. In the conventional coded OTFS system, codewords from different blocks are independent. If we further assume that the channel impulse response is independent across blocks, consistent with the wide-sense stationary uncorrelated scattering (WSSUS) assumption within the stationarity region [21], then, as indicated by our lower-bound analysis, the coding gain achievable by independent coded OTFS is quite limited.

At the transmitter, a portion of the encoded bits is superimposed onto the codeword of the next time slot. Formally, let $\alpha \in [0, 1]$ denote the superposition fraction. A selection matrix $\mathbf{S}$ is constructed by masking $n(1 - \alpha)$ columns of a random $n \times n$ permutation matrix, such that $\mathbf{v}\mathbf{S}$ represents an interleaved version of $\mathbf{v}$ with some bits set to zero. The transmitted block at time t is given by

$$\mathbf{c}^{(t)} = \mathbf{v}^{(t)} \oplus \mathbf{v}^{(t-1)}\mathbf{S}, \quad (2)$$

which is then mapped and modulated for OTFS transmission. The overall procedure is summarized in Algorithm 1 and illustrated in Fig. 4. Note that $\oplus$ denotes binary addition, i.e., the superposition is performed over $\mathbb{F}_2$, in contrast to the real-valued superposition in [22].

At the receiver, after OTFS demodulation and SPA detection, the LLRs of $\mathbf{y}^{(t)}$ are obtained. A sliding-window decoding (SWD) with window size d is then applied, modeled as iterative message passing over a normal graph with d layers [11], as shown in Fig. 4. Each layer consists of six types of nodes: the node **OTFS** and node **M** perform OTFS modulation/demodulation and mapping/demapping; the node **+** and node **=** perform as check and variable nodes in LDPC codes with additional half-edges for superposition; the node **S** transfers the selected fraction α of bits between **=** and

Algorithm 1 Transmission of the SC-OTFS Systems

Initialization: Set $\mathbf{v}^{(-1)} = \mathbf{0} \in \mathbb{F}_2^n$.

for $t = 0$ **to** $L - 1$ **do**

Basic Encoding: Encode $\mathbf{u}^{(t)} \in \mathbb{F}_2^k$ into $\mathbf{v}^{(t)} \in \mathbb{F}_2^n$ by the encoding algorithm of the 5G LDPC code $\mathcal{C}[n, k]$.

Superposition: Compute $\mathbf{c}^{(t)} = \mathbf{v}^{(t)} \oplus \mathbf{v}^{(t-1)}\mathbf{S}$, which is taken as the t -th transmitted codeword.

Signal Mapping: Generate the modulated symbols of the t -th block $\mathbf{x}^{(t)}$ in DD domain mapped from the coded block $\mathbf{c}^{(t)}$.

OTFS Modulation: Produce the time domain transmitted signal at the t -th time slot.

Termination: Set $\mathbf{u}^{(L)} = \mathbf{0}$ and obtain symbols $\mathbf{x}^{(t)}$ by the loop, where only $n\alpha$ active bits need to be transmitted.

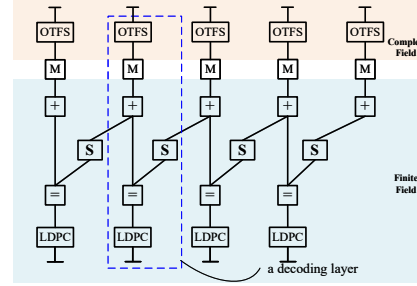


Fig. 4. The normal graph for the SC-OTFS with $L = 4$.

+; and the node **LDPC** enables the connected edges to satisfy the parity-check constraints of the 5G LDPC code. Given $(\mathbf{y}^{(t)}, \dots, \mathbf{y}^{(t+d-1)})$, the decoded block is expressed as $\hat{\mathbf{v}}^{(t)} = \mathbf{W}\mathbf{D}(\mathbf{y}^{(t)}, \dots, \mathbf{y}^{(t+d-1)})$, where $\mathbf{W}\mathbf{D}(\cdot)$ denotes the iterative window decoding with forward recursion

$$\mathbf{OTFS} \rightarrow \mathbf{M} \rightarrow \mathbf{+} \rightarrow \mathbf{=} \rightarrow \mathbf{LDPC} \rightarrow \mathbf{=} \rightarrow \mathbf{S},$$

and backward recursion

$$\mathbf{=} \rightarrow \mathbf{LDPC} \rightarrow \mathbf{=} \rightarrow \mathbf{+} \rightarrow \mathbf{S}.$$

2.3. Design Optimization by EXIT Chart Analysis

To optimize the superposition fraction α of the proposed SC-OTFS system, we adopt an EXIT chart analysis. However, due to the non-ergodic nature of OTFS-demodulated signals in high-mobility scenarios, the conventional EXIT analysis is not directly applicable. To address this, we estimate the PDF of the detector output MI (Fig. 2) and sample it to obtain the initial *a priori* MI. Specifically, L samples $I_{\text{E,Det}}^{(i)}$ ($0 \leq i \leq L - 1$) are generated and assigned to the L layers as the initial messages $I^{\text{M} \rightarrow +}$, rather than feeding them directly to node **LDPC**. The subsequent protograph-based EXIT analysis follows the update schedule of the SWD over the d -layer normal graph in Fig. 4. In the MI updates, the inputs from node **OTFS** and node **M** to node **+** are obtained via sampling; the update rules for nodes **+**, **=**, and **S** follow [13]; and node **LDPC** is updated according to the original 5G LDPC protograph. The complete procedure is summarized in Algorithm 2. This EXIT chart analysis en-

Algorithm 2 EXIT Chart Analysis for SC-OTFS

1. **Sampling:** For a given E_b/N_0 , estimate the PDF of the SPA detector output MI and randomly sample L values $\mathbf{I}_{E, \text{Det}}$.
 2. **Initialization:** Set messages at nodes $\boxed{=}$ to $\mathbf{I}_{E, \text{Det}}$, others to 0. Choose $J_{\max}$ and window size d .
 3. **Recursion:** For $t = 0, \dots, L-1$, perform MI message passing over d layers according to

$$\boxed{+} \rightarrow \boxed{S} \rightarrow \boxed{=} \rightarrow \boxed{\text{LDPC}} \rightarrow \boxed{=} \rightarrow \boxed{S} \rightarrow \boxed{+},$$
and compute a posteriori MI I_j^{APP} ($0 \leq j \leq n_b - 1$).
 4. **Output:** After $J_{\max}$ iterations, check convergence at $\boxed{\text{LDPC}}$. Decoding succeeds if all layers are recovered; otherwise, stop.
-

ables the determination of decoding thresholds for different configurations over high-mobility channels.

3. NUMERICAL RESULTS

Without loss of generality, we consider a BPSK modulation over doubly-selective channels with uniform power profile.

Example 2: Consider the SC-OTFS transmission scenario with a relative UE speed of 150 km/h and five independent resolvable paths ($P = 5$). We employ the 5G LDPC codes constructed from BG2 with a code length of 512 and a code rate of $1/2$ as the component codes. The superposition fractions range from $\alpha = 0.3$ to 0.7 with a coupling length of $L = 100$. At the receiver, the SWD with window size $d = 4$ is employed for recovery, with the SISO LDPC decoder limited to 50 iterations. Given a target FER $= 10^{-3}$, we can obtain the required E_b/N_0 by simulation. We can also obtain the decoding thresholds for different α by the proposed EXIT chart analysis. The comparison results are shown in Fig. 5, from which we observe that the trends of the simulation results are consistent with those of the analysis by the proposed EXIT chart for different superposition fractions α , and the optimized fraction is $\alpha = 0.6$ for the decoding threshold and the required E_b/N_0 . The consistency between simulation and analysis confirms the effectiveness of the study.

Example 3: Consider the OTFS transmission scenario with a relative UE speed of 150 km/h and varying numbers of independent resolvable paths of $P = 2, 3$, and 4. For the proposed SC-OTFS system, we employ the 5G LDPC codes constructed from BG2 with a code length of 512 and a code rate of $1/2$ as the component codes. The superposition fraction is $\alpha = 0.6$ and the coupling length is $L = 100$. At the receiver, the SWD with $d = 4$ is performed for recovery. For comparison, the conventional coded OTFS employs the original 5G LDPC code with length 2080 and rate $1/2$, so that the overall decoding latency is comparable to that of the SC-OTFS system. The performance under different P are shown in Fig. 6. It can be observed that: i) as the number of independent resolvable paths increases, the performance of both the conventional coded OTFS system and the SC-OTFS system can be improved, and ii) for different configurations,

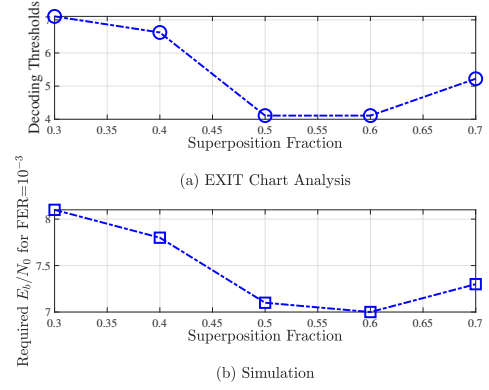


Fig. 5. The decoding thresholds by EXIT chart analysis and the required E_b/N_0 by simulations for different α .

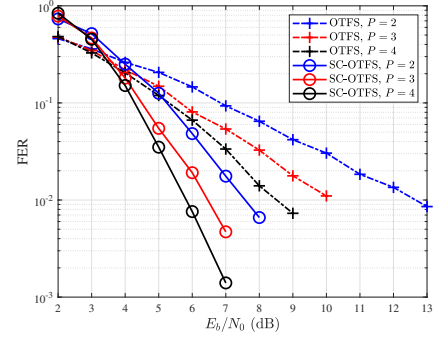


Fig. 6. The performance of the SC-OTFS and the conventional coded OTFS for different P .

the proposed schemes consistently outperform the conventional 5G LDPC coded OTFS systems, achieving more than 3 dB performance gains. Notably, we also observe that, in high E_b/N_0 region, the SC-OTFS with $n = 512$ and $P = 2$ outperforms conventional coded OTFS system with longer block length $n = 2080$ and more channel paths $P = 4$, demonstrating the advantage of the proposed SC-OTFS system.

4. CONCLUSIONS

In this paper, we proposed a novel SC-OTFS scheme by integrating the BMST framework with OTFS modulation. We analyzed the asymptotic performance of LDPC-coded OTFS and showed that increasing the code length yields only marginal gains in doubly-selective channels with limited resolvable paths. To overcome this, we designed an SC-OTFS scheme where part of each 5G LDPC-coded block is superimposed onto the subsequent block, enabling coding and inter-block diversity gains in high-mobility scenarios. An EXIT chart analysis was further developed to optimize the superposition fraction, and simulations demonstrated up to a 3 dB gain over conventional 5G LDPC-coded OTFS.

5. REFERENCES

- [1] R. Hadani, S. Rakib, M. Tsatsanis, A. Monk, A. J. Goldsmith, A. F. Molisch, and R. Calderbank, "Orthogonal time frequency space modulation," in *2017 IEEE Wireless Communications and Networking Conference*, 2017, pp. 1–6.
- [2] S. S. Das and R. Prasad, *OTFS: Orthogonal Time Frequency Space Modulation A Waveform for 6G*, 2021, pp. i–xxvi.
- [3] Z. Wei, W. Yuan, S. Li, J. Yuan, G. Bharatula, R. Hadani, and L. Hanzo, "Orthogonal time-frequency space modulation: A promising next-generation waveform," *IEEE Wireless Communications*, vol. 28, no. 4, pp. 136–144, 2021.
- [4] S. Li, J. Yuan, W. Yuan, Z. Wei, B. Bai, and D. W. K. Ng, "Performance analysis of coded OTFS systems over high-mobility channels," *IEEE Transactions on Wireless Communications*, vol. 20, no. 9, pp. 6033–6048, 2021.
- [5] R. M. Augustine, G. D. Surabhi, and A. Chockalingam, "Space-time coded otfs modulation in high-doppler channels," in *2019 IEEE 89th Vehicular Technology Conference*, 2019, pp. 1–6.
- [6] H. Zhang, L. Jing, N. Zhao, C. He, and Y. Li, "Low-complexity polar-coded iterative OTFS receiver," *IEEE Communications Letters*, vol. 29, no. 1, pp. 60–64, 2025.
- [7] 3GPP, Multiplexing and channel coding (Release 15), Jul. 3GPP TS 38.212 V15.1.1, 2018.
- [8] C. Zhang, W. Xing, J. Yuan, and Y. Zhou, "Performance of LDPC coded OTFS systems over high mobility channels," *ZTE Communications*, vol. 19, no. 4, pp. 45–53, 2021.
- [9] Q. Tao, T. Xie, X. Hu, S. Zhang, and D. Ding, "Channel estimation and detection for intelligent reflecting surface-assisted orthogonal time frequency space systems," *IEEE Transactions on Wireless Communications*, vol. 23, no. 8, pp. 8419–8431, 2024.
- [10] X. Ma, C. Liang, K. Huang, and Q. Zhuang, "Block Markov superposition transmission: Construction of big convolutional codes from short codes," *IEEE Transactions on Information Theory*, vol. 61, no. 6, pp. 3150–3163, Jun. 2015.
- [11] K. Huang and X. Ma, "Performance analysis of block Markov superposition transmission of short codes," *IEEE Journal on Selected Areas in Communications*, vol. 34, no. 2, pp. 362–374, Feb. 2016.
- [12] Q. Wang, K. Guo, and X. Ma, "Block markov superposition transmission of LDPC codes," *Journal of Information and Intelligence*, vol. 1, no. 2, pp. 115–133, 2023.
- [13] J. Yang, Q. Wang, Z. Tan, C. Li, and X. Ma, "Spatially coupled 5G LDPC codes via superposition," in *2025 IEEE Wireless Communications and Networking Conference*. IEEE, 2025, pp. 1–6.
- [14] Q. Wang, S. Cai, W. Lin, S. Zhao, L. Chen, and X. Ma, "Spatially coupled LDPC codes via partial superposition and their application to HARQ," *IEEE Transactions on Vehicular Technology*, vol. 70, no. 4, pp. 3493–3504, 2021.
- [15] Q. Wang, S. Cai, W. Lin, L. Chen, and X. Ma, "Spatially coupled LDPC codes via partial superposition," in *IEEE Int. Symp. Inf. Theory*, Paris, France, July 2019, pp. 2614–2618.
- [16] J. Yang, Q. Wang, Z. Tan, C. Li, and X. Ma, "Enhanced 5G LDPC codes and their HARQ scheme: Design and performance analysis," *Tsinghua Science and Technology*, vol. 31, no. 3, pp. 1578–1594, 2026.
- [17] Q. Wang, L. Chen, and X. Ma, "Chained LDPC codes via partial information coupling and partial parity superposition," in *2021 IEEE Globecom Workshops*, 2021, pp. 1–6.
- [18] F. Kschischang, B. Frey, and H.-A. Loeliger, "Factor graphs and the sum-product algorithm," *IEEE Transactions on Information Theory*, vol. 47, no. 2, pp. 498–519, 2001.
- [19] W. Yuan, N. Wu, A. Zhang, X. Huang, Y. Li, and L. Hanzo, "Iterative receiver design for FTN signaling aided sparse code multiple access," *IEEE Transactions on Wireless Communications*, vol. 19, no. 2, pp. 915–928, 2020.
- [20] S. Li, W. Yuan, Z. Wei, J. Yuan, B. Bai, D. W. K. Ng, and Y. Xie, "Hybrid MAP and PIC detection for OTFS modulation," *IEEE Transactions on Vehicular Technology*, vol. 70, no. 7, pp. 7193–7198, 2021.
- [21] F. Hlawatsch and G. Matz, *Wireless Communications over Rapidly Time-Varying Channels*. Academic Press, 2011.
- [22] Y. Xu, Z. Du, W. Yuan, S. Jia, and V. C. M. Leung, "Performance of OTFS-NOMA scheme for coordinated direct and relay transmission networks in high-mobility scenarios," *IEEE Wireless Communications Letters*, vol. 12, no. 12, pp. 2268–2272, 2023.